

PED-IA, a CDSS to support decision in pediatrics telephone triage: a crossover evaluation

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ABSTRACT

Background: Pediatric emergency departments face overcrowding, often driven by non-urgent consultations. Telephone triage, supported by clinical decision support systems (CDSSs), offers a potential solution to improve decision accuracy and reduce unnecessary visits. However, pediatric-specific CDSSs are scarce and underexplored.

Objective: This study aimed to evaluate the impact of a pediatric-specific CDSS, PED-IA, on decision-making accuracy, confidence, and response time.

Methods: PED-IA is an ontology-based CDSS featuring a rule-driven inference engine and a dynamic interface that guides practitioners through structured clinical reasoning. A crossover study was conducted with 51 practitioners who had to answer clinical cases with and without the CDSS. Decision accuracy, confidence, and response times were measured, and satisfaction was assessed through questionnaires.

Results: The CDSS significantly improved decision accuracy from 52.9 % to 76.0 % (+23.1 %, $p < 0.01$) and increased confidence levels by 0.65 points on a 10-point scale ($p < 0.01$). Residents benefited the most, with an improved accuracy (odds ratio of 3.70 [2.15, 6.36]). Response times increased by an average of 261.8 s per case ($p < 0.01$). Practitioners expressed high satisfaction, with 88.2 % finding the system useful for decision-making and 84.3 % believing it could reduce stress in clinical practice.

Conclusion: The PED-IA CDSS significantly enhances triage decision accuracy and user confidence, making it a promising system for clinical practice and medical education. Practitioners viewed the system positively and identified its long-term time-saving potential. Future works should focus on refining system ergonomics and exploring hybrid models that combine data-driven and logic-based approaches to improve usability and adaptability.

1. Introduction

Overcrowding in pediatric emergency departments is an ongoing issue [1,2], often exacerbated by non-urgent consultations driven by parental anxiety, lack of advice, or barriers to local care [3]. Telephone triage is proposed as a solution to reduce unnecessary admissions by

remotely evaluating the urgency of child's condition [4,5]. However, pediatric telephone triage presents some challenges. Assessing young patients remotely is inherently difficult due to nonspecific symptoms, which require detailed descriptions from parents who may be overwhelmed by emotional distress [4,6,7]. In this context, healthcare professionals, who may not be specifically trained in pediatrics [6], must

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make critical decisions quickly. This highlights the need for robust systems to support decision-making and ensure safe and effective triage [8–10].

Using clinical decision support systems (CDSSs) in call centers can enhance the quality and accuracy of pediatric telephone triage [11]. CDSSs assist healthcare professionals in making decisions [8], improving patient safety [9], and reducing costs by streamlining workflows [12,13]. A recent scoping review identified twelve knowledge-based and seven data-driven CDSSs for emergency telephone triage [14]. These systems have improved diagnostic accuracy [15] but faced limitations, including narrow clinical scope [16] or “black box” effect [17]. The review also found only one CDSS for pediatric triage [18], which relied on rigid decision trees and has been poorly adopted by health professionals. To overcome these limitations, we developed PED-IA, a CDSS specifically dedicated to pediatric telephone triage, which has flexibility, covers a wide range of pediatric emergencies, and offers explainable recommendations.

In the cross-over study presented here (CDSS vs. no CDSS), we aimed to determine whether the PED-IA CDSS can help practitioners improve their pediatric emergency telephone triage. Using AI-generated and expert-rated clinical scenarios, we assessed practitioners’ accuracy, confidence, and satisfaction and the potential impact of the CDSS on time to decision making (response time).

2. Objective

Here, we aimed to assess whether the PED-IA CDSS improved medical practitioners’ accuracy, confidence, satisfaction, and response time through a crossover study, with and without the CDSS.

2.1. Statement of significance

Issue	What is Already Known	What his Paper Adds
Pediatric telephone triage is challenging due to nonspecific symptoms and emotional factors, complicating decision accuracy.	Existing CDSS systems help triage in emergency contexts but are often not well adapted to pediatric needs, lacking flexibility and explainability.	Our pediatric-specific CDSS significantly enhances decision-making accuracy and user confidence in emergency pediatric telephone triage scenarios, particularly among less experienced residents

3. Methods

3.1. PED-IA CDSS conception

3.1.1. PED-IA knowledge base design

The knowledge base was developed using an ontology-inspired approach, offering key advantages over traditional relational databases in flexibility, interoperability, and support for clinical reasoning [19,20]. Despite the potential of ontologies in CDSS, no ontology-based system dedicated to pediatric emergency triage was found in the literature [14]. Existing ontologies in emergency medicine (e.g., ONTO-LURGENCES [21], HaPPy [22]) are poorly suited to telephone triage, as they rely on physiological parameters typically unavailable (i.e., remotely by phone).

To address this, we applied the methodology described by Bright et al. [23]. The domain was defined using 32 expert-validated recommendations from the local reference center, focusing on common pediatric emergency call scenarios. A review of existing medical terminologies, including SNOMED-CT and ICD-10, revealed that none could be reused because of a lack of granularity or inappropriate conceptual structure. Key concepts were extracted from recommendations, decomposed into atomic elements, and linked to clinical actions using

context-specific rules. A generalized decision-making logic was then formalized using axioms to ensure consistent application. The resulting knowledge base was custom-built using OWL in Protégé 5.6.3. and structured around three key roles: chief complaints, symptoms, and patient attributes. A pediatric expert reviewed all concepts, relationships, and rules to ensure clinical accuracy and completeness.

3.1.2. PED-IA CDSS inference motor

The reasoning engine is implemented as a Python web service that interfaces directly with an OWL 2 ontology using the owlready2 library [24]. Instead of relying on a traditional relational database, the system performs runtime reasoning over a structured semantic model, using OWL axioms and class restrictions to drive decision logic. The CDSS follows a modular architecture composed of three main layers:

The **knowledge layer** handles ontology traversal and dynamic knowledge retrieval. When a clinical concept is selected (e.g., *respiratory distress*), the system extracts its subclasses (e.g., *signs of respiratory struggle*), related concepts used for follow-up questioning (defined via the *hasRelatedQuestion* object property), and contextually linked concepts associated with potential clinical actions (via the *hasContext* property). All retrieved entities are tracked per user session and presented to the interface for iterative refinement of the clinical reasoning process.

The decision layer executes a rule-based inference model defined within the OWL ontology. Concepts are associated with one or more direct recommendations through the *hasDirectRecommendation* property, each mapped to a recommended action. The engine also evaluates compound logical constructs (e.g., AND, OR, some, only) and conditional axioms that bind clinical recommendations to the presence of specific contextual concepts (*hasContext*). If multiple recommendations apply, the system prioritizes and retains the most severe (precautionary principle). All decision computations are dynamically re-evaluated upon each change in concept selection.

The **presentation layer** is a web frontend that displays dynamically retrieved subclasses, related interrogation concepts, and action-linked contextual concepts based on the reasoning engine’s current state. The interface prevents redundant concept suggestions and allows users to review or deselect previously chosen concepts. Recommendation output is updated in real time to reflect the highest-severity decision consistent with the current clinical context.

3.1.3. PED-IA CDSS interface

From the user’s point of view, the PED-IA CDSS offers an interactive and structured interface that supports clinical triage in a way that mimics intuitive reasoning (Fig. 1). The process begins with selecting a chief complaint, which opens a new interface displaying three categories of information: more specific clinical signs (subclasses), related symptoms (concepts) to investigate through questioning, and contextual elements that may influence clinical action. When the user selects any of these elements, the system dynamically retrieves and displays, in turn, the same three categories for the chosen new concept, expanding the reasoning tree directly on the same page. This recursive mechanism creates an evolving branching structure that guides the user through a progressive and comprehensive case exploration. All selected concepts are continuously visible in a dedicated panel, allowing users to track and adjust their reasoning path in real time. At each step, the system updates the recommended course of action in real time, always displaying the most severe recommendation supported by the current set of inputs.

The system’s demo version, source code, and documentation are publicly available online (see Availability section).

3.1.4. PED-IA internal validation

Internal validation of the PED-IA CDSS was conducted using 96 synthetic pediatric triage cases generated with GPT-4.0. Each case represented a range of clinical scenarios and severity levels and was reviewed and refined by a pediatric expert to ensure clinical realism (e.

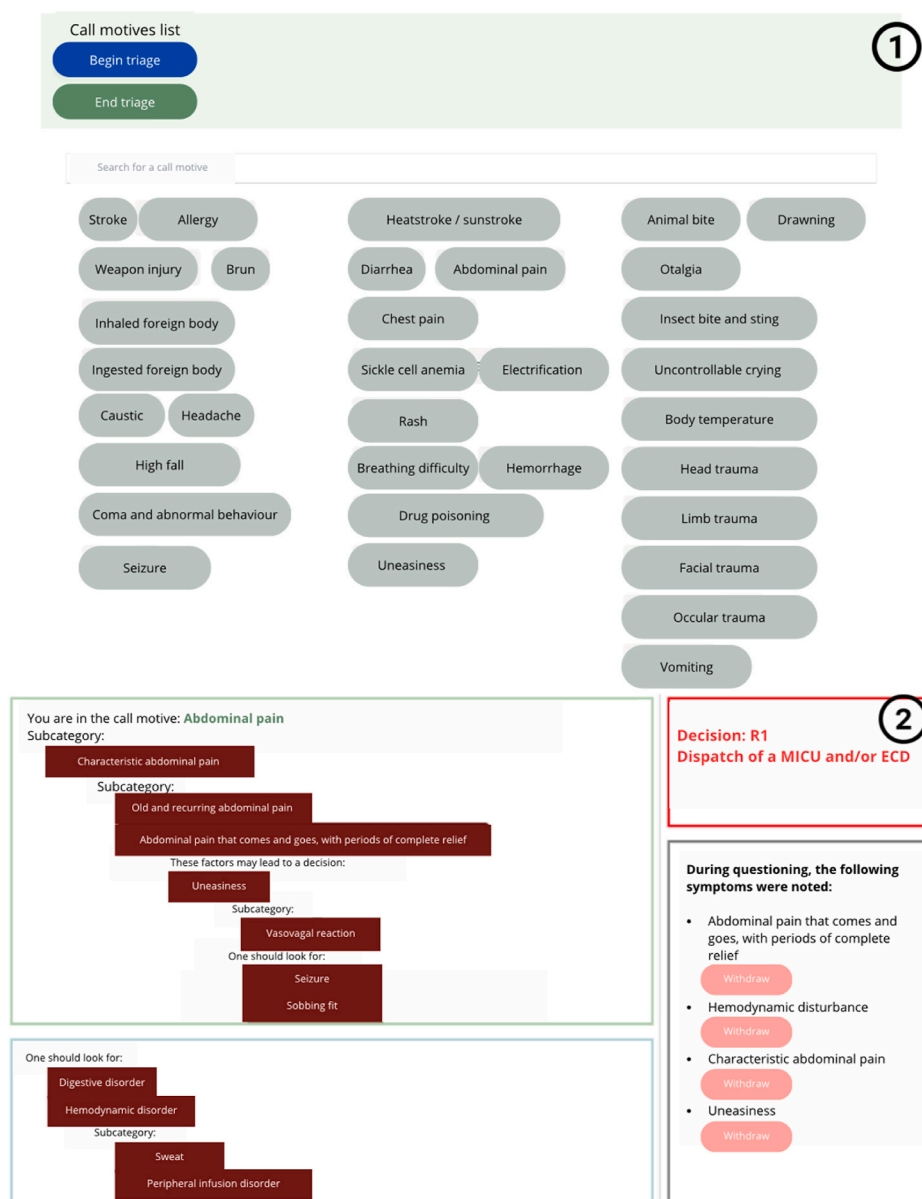


Fig. 1. PED-IA CDSS interface – translated from the screenshot. (MICU: mobile intensive care unit; ECD: emergency corresponding doctor).

g., symptom adjustments). The system's recommendations were compared to the standardized actions described in the local reference center's written recommendations. This comparison assessed the system's ability to replicate expected decisions. Across all 96 cases, the CDSS produced recommendations that were 100 % consistent with the recommendations from the local reference center, demonstrating technical validity.

3.2. PED-IA CDSS evaluation

A two-stage crossover study compared medical practitioners' accuracy, confidence, and response time to clinical cases with and without the CDSS.

This research was approved by the CER AHP Centre Research Ethics Committee (IRB: IORG0010044).

3.2.1. Practitioners' recruitment

Practitioners from several local hospitals were informally approached within clinical settings and invited to participate in the evaluation. Inclusion criteria included medical residents and/or young

medical practitioners, as they were the potential future users of the CDSS.

3.2.2. Study design

Practitioners were randomly assigned to either group A or group B. They completed a brief socio-demographic questionnaire and then answered two sets of clinical cases, with and without the CDSS (Fig. 2).

In Group A, practitioners first started to answer a set X of clinical cases without the CDSS. Then, after having completed a tutorial on the CDSS, they had to respond to a set Y of clinical cases with the CDSS.

In Group B, practitioners followed the reverse sequence. They first started with the tutorial and set X with the CDSS and then had to answer set Y of clinical cases without the CDSS.

For each case, participants reported the triage decision and their confidence level in their decision. At the end of the evaluation, participants filled in a satisfaction questionnaire regarding the CDSS. A timer was used to record the time it took to complete each clinical case set (excluding the training cases).

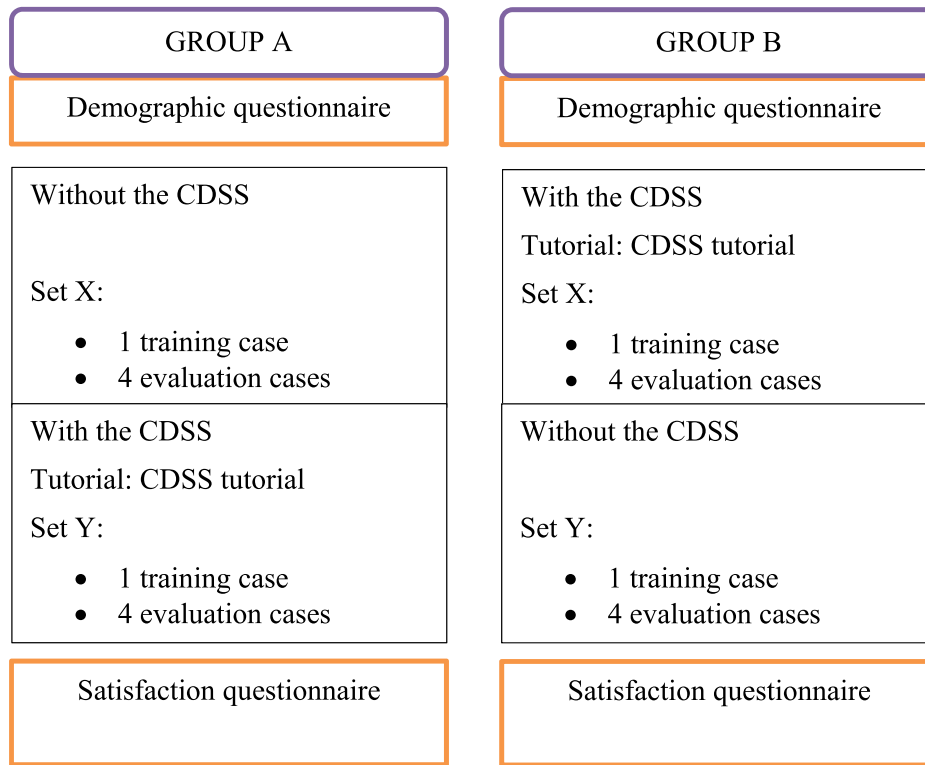


Fig. 2. Crossover study design (CDSS: Clinical Decision Support System).

3.2.3. Design of clinical cases

The two sets (X and Y) of clinical cases were randomly selected from the 96 fictitious clinical cases that were created during internal validation (See section 3.1.4). They were then reviewed and edited by a medical expert to ensure they reflected common situations in pediatric emergency settings (e.g., falls during sports, fever at school), using language typically employed by caregivers (e.g., parents or teachers) and the level of detail they would give. All cases assumed caregivers were fluent in French, and no communication barriers were introduced. The gold standard triage decision was established for each case based on the local written recommendations for emergency pediatric telephone triage.

The random selection from the 96 clinical cases led to 10 clinical cases that were then paired based on decision severity and case complexity. This process aimed to ensure that the two sets, X and Y, were comparable. This resulted in two sets of clinical cases (X and Y), each comprising 1 training case (i.e., not used for analysis) and 4 evaluation cases (i.e., used for analysis).

3.2.4. Measurement criteria

The measurement criteria were.

- Socio-demographic data, i.e., age, sex, function (resident or young practitioner), year of practice, and medical specialty;
- Triage decision accuracy for each clinical case. Triage decision accuracy was assessed by comparing each physician's selected triage level (R1, R2A, R2B, R3, or R4, as defined in Table 1) to the gold standard (as defined in section 3.2.3). A triage decision was considered: 'accurate' when it matched precisely to the recommended level; 'lower' when the assigned triage level was less urgent than recommended; and 'higher' when it was more urgent than recommended.
- Level of confidence, rated with a numeric scale ranging from 0 to 10 for each clinical case;
- Response time for each set of cases (excluding the training case);

Table 1

Triage decision variables (ECD: emergency corresponding doctor; EMS: emergency medical service; GP: general practitioner; MICU: mobile intensive care unit).

Triage decision labels	Emergency severity degree	Means to trigger
R1	Patent or latent vital emergency	Dispatch of a MICU and/or ECD
R2A	Potential emergency	Dispatch of a fire and rescue service within an appropriate timeframe
R2B	Potential emergency	EMS by own facilities or private ambulance service
R3	Relative or perceived emergency	Referral to a GP or other healthcare provider
R4	No emergency	Medical advice or medication prescription

- Satisfaction regarding the CDSS was assessed using questions about ease of use, usefulness for case resolution, utility for real-life decision-making, potential impact on stress reduction, potential time-saving benefits, and likelihood of recommending the system to colleagues. Each statement was rated on a five-point Likert scale from "Strongly disagree" to "Strongly agree";
- Free comments.

3.2.5. Statistical analysis

Generalized linear mixed models (GLMMs) were applied to evaluate the effect of CDSS usage on decision accuracy, accounting for individual variability. Comparisons of accuracy, under-triage, and over-triage rates between CDSS conditions were performed using Chi-square tests. Interaction effects, sequence effects, and the mediating role of confidence were also explored. Paired t-tests assessed differences in decision-making time between the two conditions. For satisfaction data, frequencies of responses to Likert scale items were calculated and stratified by function (resident or young practitioner). A qualitative analysis of free comments was also conducted. All statistical analyses were

performed using R software, and significance was set at $p < 0.05$.

4. Results

4.1. Practitioners' characteristics

51 practitioners were included in the study, with a mean age of 28.25 years ($SD = 1.97$) and a majority of women (72.5 %) (Table 2). Most participants were residents (84.3 %), especially from general medicine (29.4 %) and paediatrics (41.2 %).

4.2. Triage decision accuracy and level of confidence

4.2.1. Sequence and carry-over effects

To assess the crossover design's validity, we investigated whether there was an interaction between **decision** support and **period** (the order in which the CDSS was used). The interaction term in the mixed-effects logistic regression model was not statistically significant, indicating that the CDSS's effect on decision accuracy did not depend on the order of CDSS exposure.

Despite the absence of an interaction effect, a significant **period effect** was observed ($OR = 0.63$ [0.41, 0.96]). This finding indicates that the overall accuracy of decisions differed between the first and second periods. Factors such as differences in case difficulty or participant fatigue could explain this sequence effect. To account for potential differences in case complexity, the variable *case_id* was included as a random effect in the model. This adjustment allowed us to control for case-specific variability in decision difficulty.

4.2.2. Triage decision accuracy

The triage decision accuracy improved significantly from 52.9 % without the CDSS to 76.0 % with the CDSS (Table 3).

A stratified analysis showed that for residents, CDSS significantly improved decision accuracy (51.2 % versus 78.5 %, $OR = 3.70$, [2.15, 6.36]), showing that they benefitted from using the system. Conversely, for young practitioners, CDSS did not significantly affect decision accuracy (62.5 % versus 62.5 %). The adjusted odds ratio ($OR = 1.39$, [0.32, 5.99]) indicates a trend toward a positive effect, but this result is not statistically significant. The lack of significance in this group could be due to the smaller sample size (64 observations, 8 participants) and variability in experience levels. This difference was confirmed in the global mixed-effects model, which showed a significant interaction between decision support and function ($p = 0.03$). No other significant interaction was found, including with medical specialty.

Despite the use of CDSS, 24 % of errors were still present, corresponding to "lower" or "higher" triage decisions. The in-depth analysis showed that with CDSS, "lower" errors decreased from 21.6 % to 13.2 %

Table 3

Final Mixed-Effects Model: the effect of the CDSS on decision accuracy, stratified by function, i.e., residents or young practitioners (CDSS = Clinical Decision Support System).

Variable	With vs without CDSS	OR [95 %CI] (Residents)	OR [95 % CI] (Young practitioners)
Triage decision accuracy	76.0 % vs 52.9 %	3.70 [2.15, 6.36]	1.39 [0.32, 5.99]
Confidence	Mean of level: 8.80 vs 8.15	1.40 [1.16, 1.68]	1.33 [0.70, 2.51]

($p = 0.03$) while "higher" errors decreased from 25.5 % to 10.8 % ($p < 0.01$), highlighting the positive effect of the CDSS on both kind of errors (Table 4). Moreover, the analysis of these 24 % errors showed that for 17.6 % of them, the triage decision severity differed by one level compared to the gold standard (e.g., the practitioner chose R3, instead of R4).

4.2.3. Level of confidence and mediation analysis

The mean confidence level improved significantly from 8.15 to 8.80 with the CDSS, i.e., having the CDSS significantly increases confidence by 0.65 points ($p < 0.01$). Meanwhile, being a resident compared to being a young practitioner does not significantly impact confidence.

With the CDSS, participants tend to have higher confidence, especially when they gave correct responses (Fig. 3).

The mediation analysis revealed that the CDSS increased the odds ratio of a correct response by 26.0 % ($OR = 1.26$ [1.16, 1.36]). Of this effect, 4.0 % was mediated through increased confidence (Average Causal Mediated Effect (ACME): $OR = 1.04$ [1.02, 1.07]), while the direct effect of decision support on accuracy was 22.0 % (Average Direct Effects (ADE): $OR = 1.22$ [1.13, 1.32]).

4.3. Response time

The mean response time was significantly higher, from 132.9 s to 394.7 s ($p < 0.01$) with the CDSS (Table 5).

4.4. Participant satisfaction

Overall, practitioners seemed to have a generally positive attitude towards the CDSS (Fig. 4).

78.4 % of them ($n = 40$) found the system easy to use; and 90.2 % ($n = 46$) found the CDSS useful for case resolution. Additionally, 88.2 % ($n = 45$) stated that the CDSS would be useful for telephone triage decision-making, 70.6 % ($n = 36$) thought it would save time, and 84.3 % ($n = 43$) thought it would help reduce stress. 88.2 % ($n = 45$) of practitioners were willing to recommend the system to their colleagues.

Open comments mainly were positive, highlighting the system's ease of use and potential usefulness. However, some issues were noted, such as the need to get used to the system to use it more effectively. A few participants suggested improvements, such as enhancing usability and adding specific clinical recommendations.

5. Discussion

This study evaluated the impact of the PED-IA CDSS on decision-making abilities in pediatric emergency telephone triage. The results demonstrated that the CDSS significantly increased both decision accuracy (+23.1 %, $OR = 3.70$ [2.15, 6.36]) and practitioner confidence (+0.65 points, $OR = 1.40$ [1.16, 1.68]). However, these benefits were accompanied by increased response times (+261.8 s, $p < 0.01$). Overall, practitioners expressed positive attitudes towards the CDSS, emphasizing its ease of use and potential utility in clinical practice and training.

Table 2
Practitioners' characteristics.

Characteristics	Overall (n = 51)
Gender	
Women	37 (72.5 %)
Men	14 (27.5 %)
Age (mean $\pm$ SD)	28.25 $\pm$ 1.97
Specialty	
Paediatrics	21 (41.2 %)
General Medicine	15 (29.4 %)
Emergency Medicine	10 (19.6 %)
Anesthesiology	1 (2.0 %)
Dermatology	1 (2.0 %)
Nephrology	1 (2.0 %)
Neurology	1 (2.0 %)
Psychiatry	1 (2.0 %)
Function	
Resident	43 (84.3 %)
Young practitioner	8 (15.7 %)

Table 4
Analysis of errors made with the PED-IA CDSS. Lower = underestimation of severity in the practitioner’s decision compared to the gold standard (e.g., dispatching a fire and rescue service [R2A] instead of a mobile intensive care unit [R1]); Higher = overestimation of severity (e.g., referring a patient to a general practitioner [R3] instead of simple medical advice [R4]). (CDSS = Clinical Decision Support System).

	WITHOUT CDSS				WITH CDSS			
	Cases	Accurate	Lower	Higher	Cases	Accurate	Lower	Higher
GROUP A	Set X	56.7 %	13.5 %	29.8 %	Set Y	69.2 %	14.4 %	16.3 %
GROUP B	Set Y	49 %	30 %	21 %	Set X	83 %	12 %	5 %
Total		52.9 %	21.6 %	25.5 %		76.0 %	13.2 %	10.8 %
p-value	(without VS with)					< 0.01	0.03	<0.01

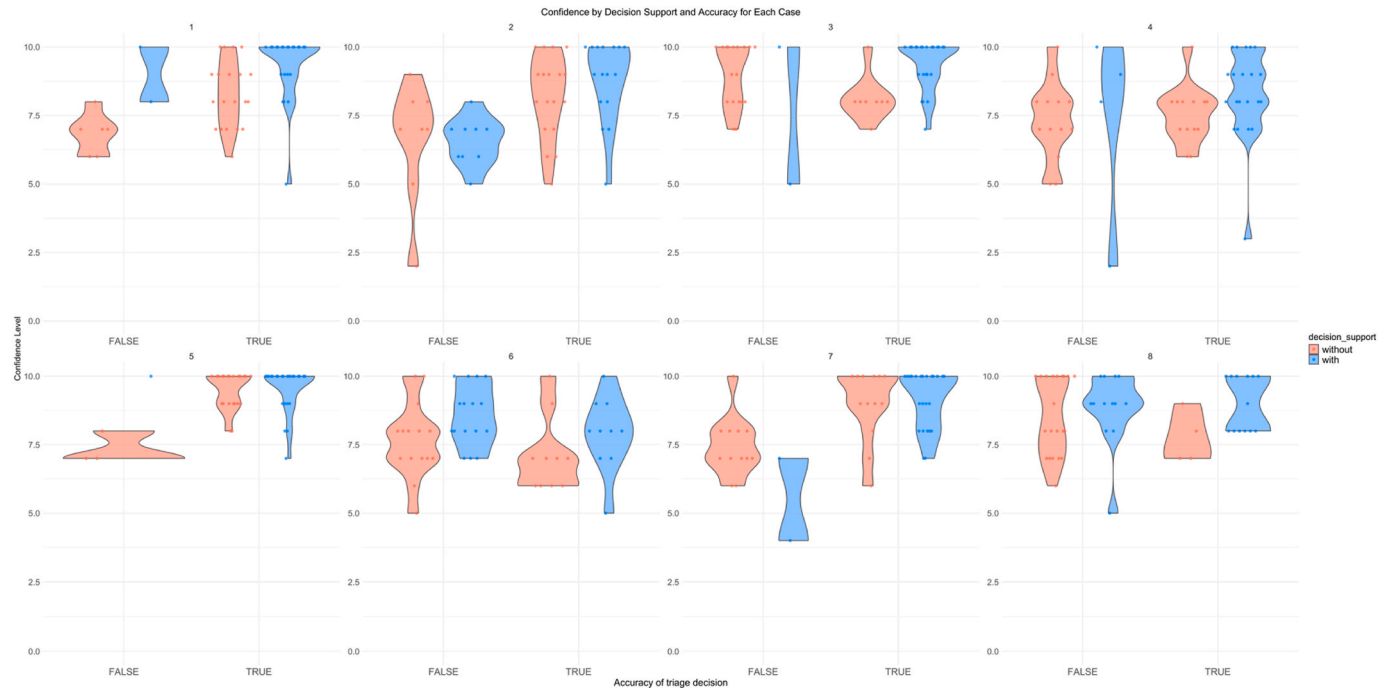


Fig. 3. Confidence levels by case (case 1 to case 8) and Clinical Decision Support System (CDSS) usage. Each plot shows the distribution of confidence scores (0–10) for responses without the CDSS (red) and with the CDSS (blue), when the participant was aligned with the gold standard (TRUE) or not (FALSE). The width of each violin represents the frequency of responses at each score. Overall, confidence levels tend to increase when using the CDSS. However, confidence remains variable in some cases (e.g., 2 and 6), suggesting persistent uncertainty or complex interpretation despite decision support. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 5
Paired t-test results for mean response times with and without the Clinical Decision Support System (CDSS).

Group	Time with the CDSS (seconds)	Time without the CDSS (seconds)	Mean Difference (seconds)	95 % CI	p-value
Group A	387.6	151.2	236.44	[150.13; 322.74]	<0.01
Group B	402.1	113.9	288.20	[206.42; 369.99]	<0.01
Combined Sample	394.7	132.9	261.81	[203.93; 319.70]	<0.01

5.1. Limitations and strengths

Several limitations should be acknowledged. First, the study’s sample size, particularly for young practitioners, was relatively small, which may limit the generalizability of the findings, even if the crossover study design allowed to demonstrate the benefits with a reduced number of practitioners [25].

Likewise, a selection bias may have occurred, as participants who chose to participate in the study might be more inclined to adopt new technologies or more motivated to improve their skills. The non-participation of experienced pediatric triage professionals limits the possibility of generalization but is in line with the study’s aim to evaluate the CDSS with potential future system users. Moreover, although carefully designed and validated, clinical cases may not fully capture the complexity and unpredictability of real-world scenarios. Additionally, the controlled study environment does not account for external factors that may influence decision-making in actual clinical settings, such as multitasking, interruptions, or emotional stress, or communication barriers like language difficulties or limited health literacy [26]. Nevertheless, matching clinical cases by severity and complexity ensured a fair comparison between conditions with and without the CDSS. Furthermore, the study focuses on accuracy, confidence, satisfaction, and response time to comprehensively evaluate the CDSS. Additional criteria, such as usability score [27,28] or stress levels, should be considered in future evaluations, could provide more in-depth insights. Finally, this evaluation represents the first stage in the software’s lifecycle, providing preliminary feedback before broader clinical trials.

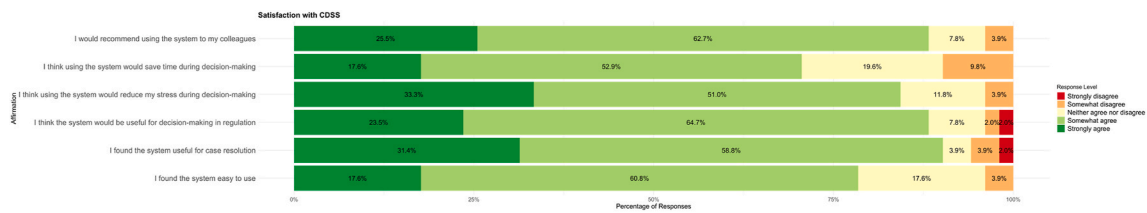


Fig. 4. Satisfaction of practitioners with the Clinical Decision Support System (CDSS).

The absence of significant carry-over effects supports the robustness of the design. Still, the period effect observed suggests potential fatigue or variability in case difficulty, which could be mitigated in future studies. Additionally, as the study was conducted in a simulated setting, it does not allow for the evaluation of real-life clinical outcomes, such as missed diagnoses or unnecessary referrals. Future studies in real-world settings will be needed to assess the CDSS's diagnostic performance, particularly its sensitivity, specificity, and impact on emergency care [14].

5.2. Interpretation

The CDSS significantly improved decision accuracy, particularly among residents. The observed improvement from 52.9 % to 76.0 % reflects the system's ability to assist less experienced practitioners. These findings are consistent with previous research highlighting the potential of decision support systems to enhance diagnostic accuracy and patient safety in emergency settings [29,30]. However, the system had no significant impact on the accuracy of young practitioners, possibly due to their higher baseline performance and smaller sample size. This finding suggests that less experienced clinicians may benefit more from decision support systems, perhaps because they need more guidance in their clinical reasoning [31,32], especially in specialized areas of medicine such as pediatric emergency triage. Errors primarily involved minor deviations from the gold standard (one level over), highlighting the challenges associated with nuanced triage decisions. Indeed, pediatric telephone triage is a field that remains poorly documented and lacks comprehensive national and international guidelines [29], with decisions often relying on personal experience and local practice [33,34]. These factors contribute to the difficulty of aligning expert decisions with a gold standard [35], underlining the complexity of making consistent triage decisions.

Moreover, the distinction between "lower" and "higher" errors provides important clinical insights. Both decreased with CDSS use. This suggests that the system has led to an improvement in overall triage accuracy. The analysis of the different error categories reveals that both, patient-endangering clinical "under-triage" and unnecessary over-utilization of emergency resources by "over-triage" were reduced through the use of the CDSS in test scenarios. Two main hypotheses can explain these residual errors: either the CDSS provided an appropriate recommendation that was voluntarily not followed, or practitioners entered incomplete or inaccurate clinical data when using the CDSS. These results highlight the importance of optimizing system usability, ensuring proper training, and designing interfaces that promote accurate and complete data capture.

Additionally, while confidence increased significantly with the CDSS (+0.65 points), participants' baseline confidence was already high, which is surprising in this highly specific field. Confidence in the system itself may play a critical role in its adoption, which the integration of explanation features could enhance [27,28,36]. For example, highlighting the reasoning pathways that lead to specific decisions could improve transparency and explainability, making it easier for users to understand and trust the recommendations provided.

Finally, the CDSS increased response times significantly by around 1 min per case. This result was expected, as participants required time to familiarize themselves with the system. Improving the system's

ergonomics—for example, by incorporating icons, tables, graphics, or visual aids—could make navigation faster and more intuitive [37]. The balance between time spent and gains in accuracy is crucial for real-world adoption, as systems that take too long may encounter resistance despite their benefits [38]. Moreover, as users become more familiar with the system, efficiency may improve [23,39,40], and practitioners seemed to think so, too, as they acknowledged that the system would allow them to gain time in the future in the satisfaction questionnaire.

5.3. Comparison with other CDSSs

Our pediatric-specific CDSS addresses similar variability challenges while demonstrating improved decision-making accuracy compared to logic-based decision support systems for emergency telephone triage. For instance, the After Hours Telephone Care Program [18], a pediatric-focused system, reported 81 % agreement between nurse dispositions and protocol recommendations and an interrater reliability κ of 0.46. Variability was due to incomplete adherence to the protocol or missing information, highlighting the dependence of such systems on user experience and consistent application of guidelines.

Similarly, the NHS Direct system, designed for adult emergencies, revealed significant variability in triage decisions depending on the type of software used and nurses' experience, with self-care recommendations ranging from 31 % to 44 % [41].

Unlike these studies conducted in real-life settings, our evaluation was based on fictional cases, which may reduce external influences [42] such as user stress or incomplete information. Our study showed that the PED-IA CDSS improved decision accuracy from 53 % to 76 % when support was provided, with most errors occurring within one level of the gold standard. These results are consistent with findings in logic-based systems, especially in pediatric contexts, where decision-making is shaped by adherence to protocols and dispatchers' habits.

Compared to data-driven systems, our CDSS balances predictive performance with explainability and user interaction. For instance, DeepEMC2 [17], a deep ensemble multitask model developed for emergency medical call classification, achieved macro F1-scores of 0.759 for life-threatening classification, representing a significant performance improvement over baseline machine learning models and current protocols. Similarly, the Yokohama New Emergency System [43], a data-driven model designed for adult emergencies, achieved high sensitivity (80 %) and specificity (96 %) in identifying life-threatening conditions. However, its performance varied based on the type of caller and data quality. Both systems illustrated the strength of data-driven approaches in extracting and processing complex, multimodal data to enhance decision accuracy.

In contrast, our CDSS emphasizes explainability and direct user interaction rather than predictive complexity. The variability observed in our system predominantly aligns with the findings from DeepEMC2, suggesting that unconsidered information, such as communication nuances, significantly impacts decision-making. Moreover, DeepEMC2's dependence on free-text data highlights the challenges of extracting subjective information. Future efforts should further focus on developing a hybrid system that combines the strengths of data-driven models, such as capturing complex patterns, with the transparency

and explainability of logic-based systems to enhance accuracy and adaptability in pediatric emergency triage [44,45].

5.4. Perspectives

5.4.1. Concerning the CDSS design

The current prototype uses symbolic artificial intelligence with a knowledge base. Plans include enriching the knowledge base with new call reasons organized around all physiological systems and patient-friendly vocabulary to improve communication with patients and enhance care quality [46]. We also aim to integrate machine learning capabilities to develop a hybrid system that leverages both logical decision frameworks and data-driven techniques [47]. This could improve accuracy [47], address recommendation gaps [48], and adapt to various clinical scenarios [49]. The success of this system depends on access to high-quality pediatric triage data, which is a challenge due to privacy and standardization issues [50]. Integration with local emergency systems would enable dispatchers to access real-time resource availability, optimizing decision-making and response efforts [42,51].

5.4.2. Concerning its implementation

The findings suggest integrating the CDSS into resident training programs could benefit greatly. As residents appeared to gain the most from the system, incorporating it into educational curricula may enhance their clinical reasoning skills and decision-making confidence in pediatric emergencies. The CDSS could serve as a teaching e-learning system that provides immediate decision support and facilitates learning through its explainable recommendations and iterative inquiry process. For instance, an AI-powered CDSS platform for radiology trainees demonstrated that simulation cases with CDSS feedback significantly improved perceived educational value compared to clinical cases without the CDSS (1.8 vs. 2.2, $p < 0.05$) while maintaining or slightly reducing overall case handling time [52]. Moreover, trainees actively engaged with the system, revising their diagnoses in 44 % of cases based on CDSS feedback, illustrating how such systems can enhance learning by providing structured, real-time guidance in decision-making.

5.4.3. Concerning its evaluation

In the future, the CDSS should be evaluated iteratively throughout the information technology lifecycle [53,54]: (i) during development with real-world data [55], (ii) during pre-implementation with prospective datasets [56], (iii) during implementation through randomized trials [57], and (iv) during follow-up for ongoing monitoring [53,56]. Evaluation should include both intrinsic system characteristics (e.g., safety, reliability) and clinical impact (e.g., mortality, cost-effectiveness, user satisfaction) [14]. In the context of using the CDSS to train residents, future studies could explore the long-term impact of CDSS use on residents' skills and autonomy in clinical practice [58,59]. Assessing whether prolonged use leads to improved efficiency, reduced reliance on the system, and better patient outcomes would be valuable [23,60].

6. Conclusion

This study demonstrates that our pediatric-specific CDSS significantly enhances decision-making accuracy and user confidence in emergency pediatric telephone triage scenarios, particularly among less experienced residents. Although the system increased response times, practitioners highlighted its ease of use and potential to improve triage quality and reduce stress. These findings underscore the CDSS's dual role as both a practical decision-making aid and a valuable educational system, suggesting that its integration into clinical practice and training programs could enhance outcomes in pediatric emergency care. Future work should optimize system ergonomics, address variability, and explore hybrid models that combine data-driven insights with transparent, logic-based frameworks to improve adaptability and effectiveness.

CRedit authorship contribution statement

Aurélia Manns: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Alix Millet:** Methodology, Investigation, Data curation, Conceptualization. **Fleur Mougin:** Writing – review & editing, Validation, Supervision, Software. **Florence Campeotto:** Writing – review & editing, Validation, Supervision, Resources, Conceptualization. **Benoît Vivien:** Validation, Supervision, Resources, Conceptualization. **Laurent Dupic:** Validation, Supervision, Resources, Conceptualization. **Anita Burgun:** Writing – review & editing, Validation, Supervision, Resources, Conceptualization. **Jean-Philippe Jais:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Formal analysis. **Rosy Tsopra:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation, Conceptualization.

Ethical approval

This research was approved by the CER AP-HP—Centre Research Ethics Committee (IRB: IORG0010044).

Availability of data, materials, and code

A demo version of the CDSS is publicly available at: <https://prod.d35d12lv6zuvjs.amplifyapp.com/>

The source code and dataset used for demonstration purposes can be accessed on GitHub: <https://github.com/auremns/PED-IA>.

All materials are provided under an open-source license for non-commercial research and educational use.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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